Statistical Analysis

(Basics)

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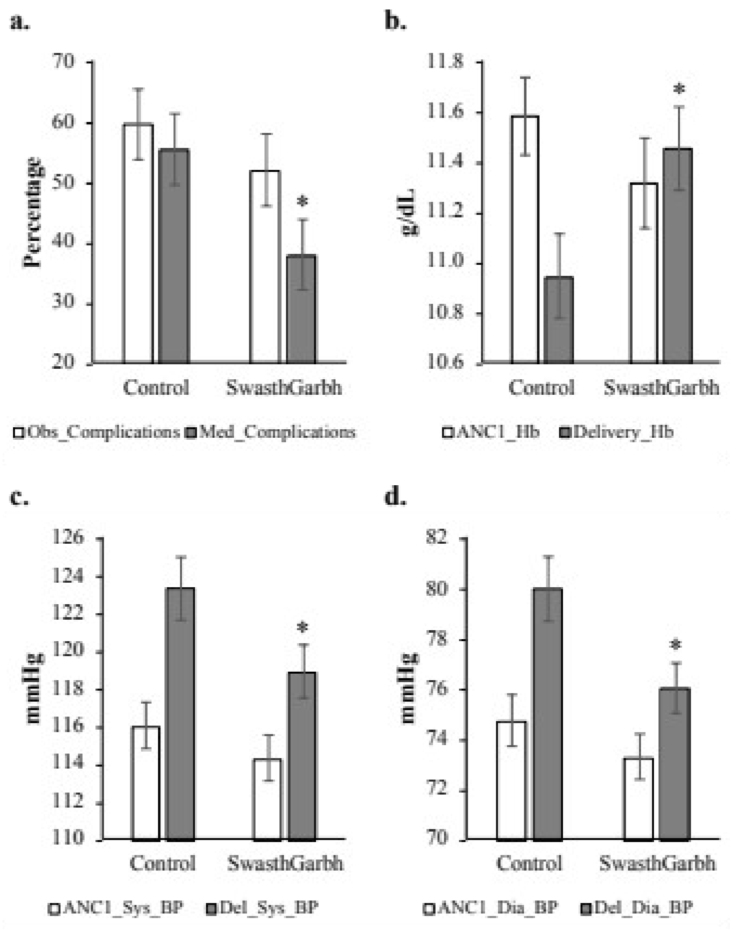
Need to study statistics

How will we prove that SwasthGarbh App is useful?



Baseline categorical and continuous variables were summarized as frequency (%) and mean ± SD, respectively. The continuous outcome variables that followed normal distribution were compared between the groups using unpaired t-test and those not following normal distribution were compared between the groups using the Wilcoxon rank sum test. The paired t-test/signed rank test was used to compare the change from baseline. Categorical variables were compared between the groups using Chi-square/Fisher's exact test. Linear regression analysis was done to find the effect of the treatment on the number of antenatal visits adjusting for socio-demographic variables. A two-sided p-value was considered statistically significant. All the statistical analysis was carried out using Stata 15.0 (StataCorp LP, Texas, USA).

Fig. 6. Improvement of clinical parameters in patients registered on SwasthGarbh App (test group; $n=71$) in comparison to Control patients (not registered on App; $n=72$). (a) Obstetric Complications (Obs_Complications) and Medical Complications (Med_Complications), (b) ANC1 hemoglobin (ANC1_Hb) and Delivery hemoglobin (Delivery_Hb), (c) ANC1 systolic BP (ANC1_Sys_BP) and Delivery systolic BP (Del_Sys_BP), and (d) ANC1 diastolic BP (ANC1_Dia_BP) and Delivery diastolic BP (Del_Dia_BP) [*indicates significant differences ($P<0.05$) between Medical Complications in SwasthGarbh and Control groups, or ANC1 and Delivery parameters; error bars represent standard error].

Sharma S, Soni S, … **Sharma D*** (2023) *IEEE JBHI*, 27, 2729-2738.

Need to study statistics

How will we prove any experimental result is significant?

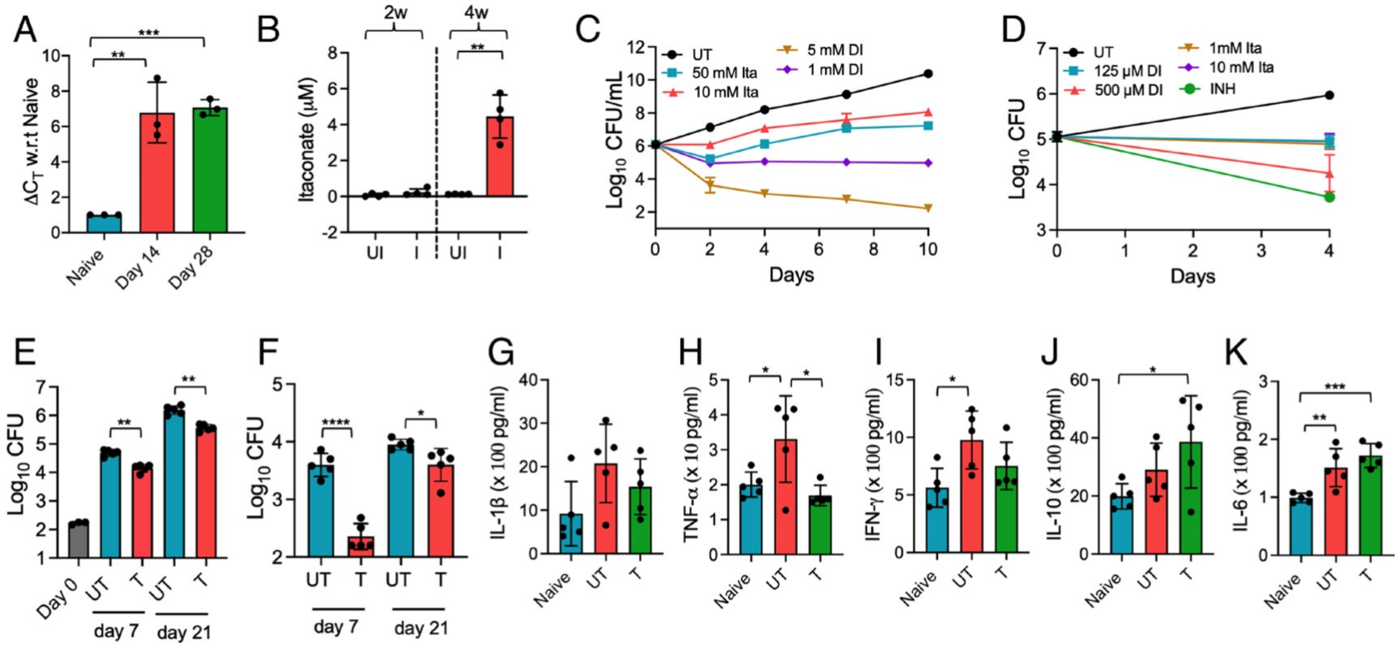


Fig. 1. Itaconate restricts the growth of *Mycobacterium tuberculosis* in vitro and in vivo. (*A* and *B*) The relative levels of *Irg1* (*A*) and itaconate (*B*) in the lung tissues of mice infected with *M. tuberculosis* at 2- and 4-wk postinfection were calculated as described in *Materials and Methods*. The data shown in panels (*A*) and (*B*) are obtained from three or four mice, respectively. (*C*) The growth of *M. tuberculosis* in MB7H9 medium was determined after exposure to various concentrations of sodium itaconate (ITA) or dimethyl itaconate (DI). (*D*) THP-1 macrophages were infected with *M. tuberculosis* strain at 1:10 MOI and treated with different concentrations of DI or ITA for 4 d. The data shown in panels C and D are the mean ± SD of $\log_{10}$ CFU obtained from two independent experiments performed in either duplicates or triplicates. (*E* and *F*) 6 to 8 wk old female BALB/c mice were infected with *M. tuberculosis* via aerosol route. The animals were administered intraperitoneally with 50 mg/kg DI for either 7 or 21 d. The data shown in these panels are mean ± SD of $\log_{10}$ CFU of lungs (*E*) and splenic (*F*) bacillary loads obtained from five animals per group. (*G*–*K*) The intracellular levels of IL-1β (*G*), TNF-α (*H*), IFN–γ (*I*), IL-10 (*J*), and IL-6 (*K*) were determined in naïve, untreated, and DI treated groups at day 21 posttreatment by ELISA. The data shown in these panels are mean ± SD obtained from five animals. The data were statistically analyzed using one-way ANOVA (*A*, *G*, *H*, *I*, *J*, and *K*) or a two-tailed “paired” *t* test (*B*, *E*, and *F*). $^{*}P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$ and $^{****}P < 0.0001$.

Priya M, … **Sharma D** … Singh R (2025) *PNAS*, 122, e2423114122.

Introduction

Statistics is the science of collection, presentation, analysis and interpretation of numerical data.

– Croxton & Cowden

Why Statistics?

- *Helps us see patterns, trends and relationships in the data*
- *Assists in interpreting, predicting, pattern analysis and drawing conclusions*
- *Applicable across multiple fields including data science, medicine, biology, economics, social sciences, etc*

Basic Concepts

- *Descriptive statistics:* *Summarizing data (mean, median, mode, dispersion).*
- *Population:* *Entire group being studied.*
- *Sample:* *A subset of the population (the information the sample provides about the population is uncertain). Eg. avg CGPA of DAI101 based on my class sample (though the sample size is ~25%, will it be correct??)*
- *Inferential statistics:* *Making predictions or inferences about a population based on a sample. It includes using hypothesis testing to assess the validity of assumptions or claims about the population.*
- *Data Types:*

  Quantitative (continuous, discrete) vs. Qualitative (nominal, ordinal).

General overview

- *Question: Difference in BP recordings between machine and human measurements (Real/Chance) [Inferential Stats]*

- *Considerations:*
  - *How many machines should we test? [quality variation]*
  - *How many people should we test at each machine?*

[Sample size estimation]

Mean blood pressures and differences between machine and human readings at four locations

		Systolic blood pressure (mm Hg)					
		Machine		Human		Difference	
Location	Number of people	Mean	Standard deviation	Mean	Standard deviation	Mean	Standard deviation
A	98	142.5	21.0	142.0	18.1	0.5	11.2
B	84	134.1	22.5	133.6	23.2	0.5	12.1
C	98	147.9	20.3	133.9	18.3	14.0	11.7
D	62	135.4	16.7	128.5	19.0	6.9	13.6

No real difference

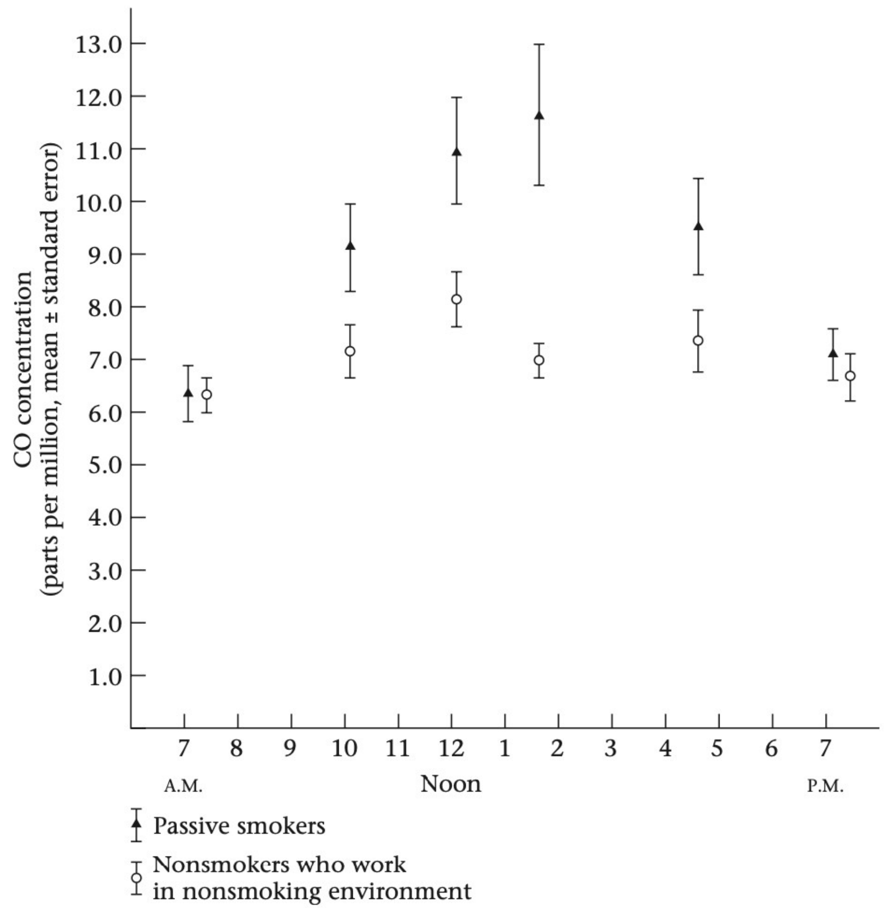
Data visualization

Graphic displays give researcher a clue to principal trends in data

- *Understandable without reading text*
- *Clear labeling*

Converge-Diverge-Converge

Mean carbon-monoxide concentration (± standard error) by time of day as measured in the working environment of passive smokers and in nonsmokers who work in a nonsmoking environment



Descriptive Statistics

Descriptive Statistics

Measures of central tendency (or location)

- Mean
- Median
- Mode

Measures of variability (or spread)

- Range
- Variance
- Standard Deviation
- IQR

Arithmatic Mean

- *Sum of all the observations divided by the number of observations. It is written in statistical terms as*

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

$$\sum_{i=1}^{n} x_i$$

is simply a short way of writing the quantity $(x_1 + x_2 + \cdots + x_n)$.

Arithmatic Mean

Sample of birthweights (g) of live-born infants born at a private hospital in San Diego, California, during a 1-week period

i	x_i	i	x_i	i	x_i	i	x_i
1	3265	6	3323	11	2581	16	2759
2	3260	7	3649	12	2841	17	3248
3	3245	8	3200	13	3609	18	3314
4	3484	9	3031	14	2838	19	3101
5	4146	10	2069	15	3541	20	2834

$$\bar{x} = (3265 + 3260 + \cdots + 2834)/20 = 3166.9 \text{ g}$$

- *Very sensitive to extreme values; hence poor measure of central tendency*
- *1st value 500g; Mean: 3028.7g*
- *Nonetheless, it is by far the most widely used measure of central tendency*

Median

- *If **n** observations are ordered from smallest to largest, then median is defined as*

(1) The $\left(\frac{n+1}{2}\right)$th largest observation if n is odd

(2) The average of the $\left(\frac{n}{2}\right)$th and $\left(\frac{n}{2}+1\right)$th largest observations if n is even

Median of the previous question

First, arrange the sample in ascending order:

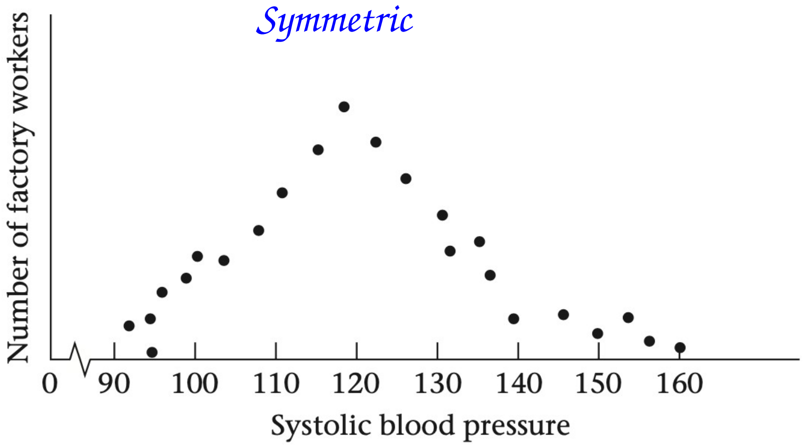
2069, 2581, 2759, 2834, 2838, 2841, 3031, 3101, 3200, 3245, 3248, 3260, 3265, 3314, 3323, 3484, 3541, 3609, 3649, 4146

Because n is even,

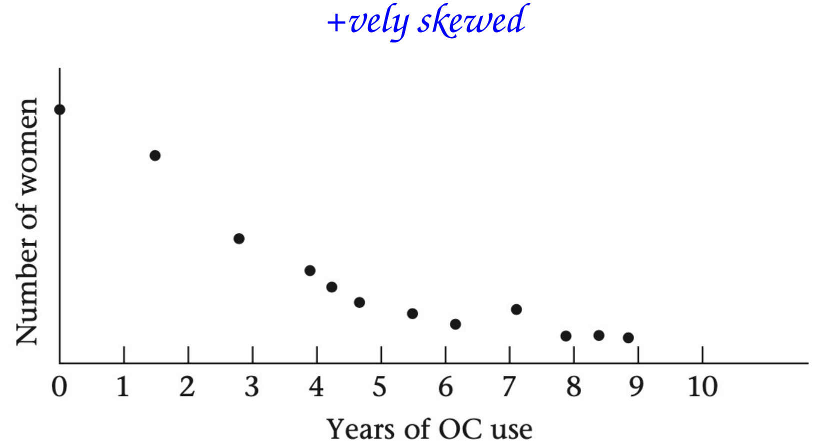
1st value 500g; Median: No change

Sample median = average of the 10th and 11th largest observations
= (3245 + 3248)/2 = 3246.5 g

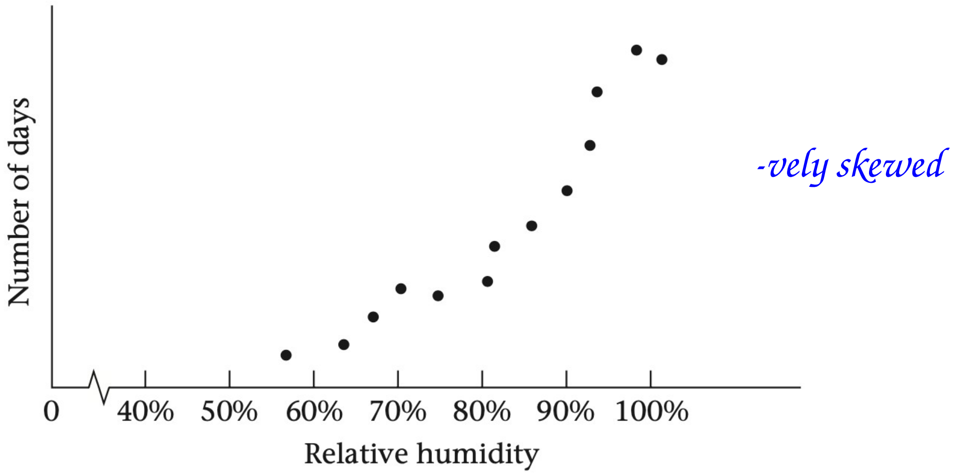
Symmetric and skewed distributions



(a)



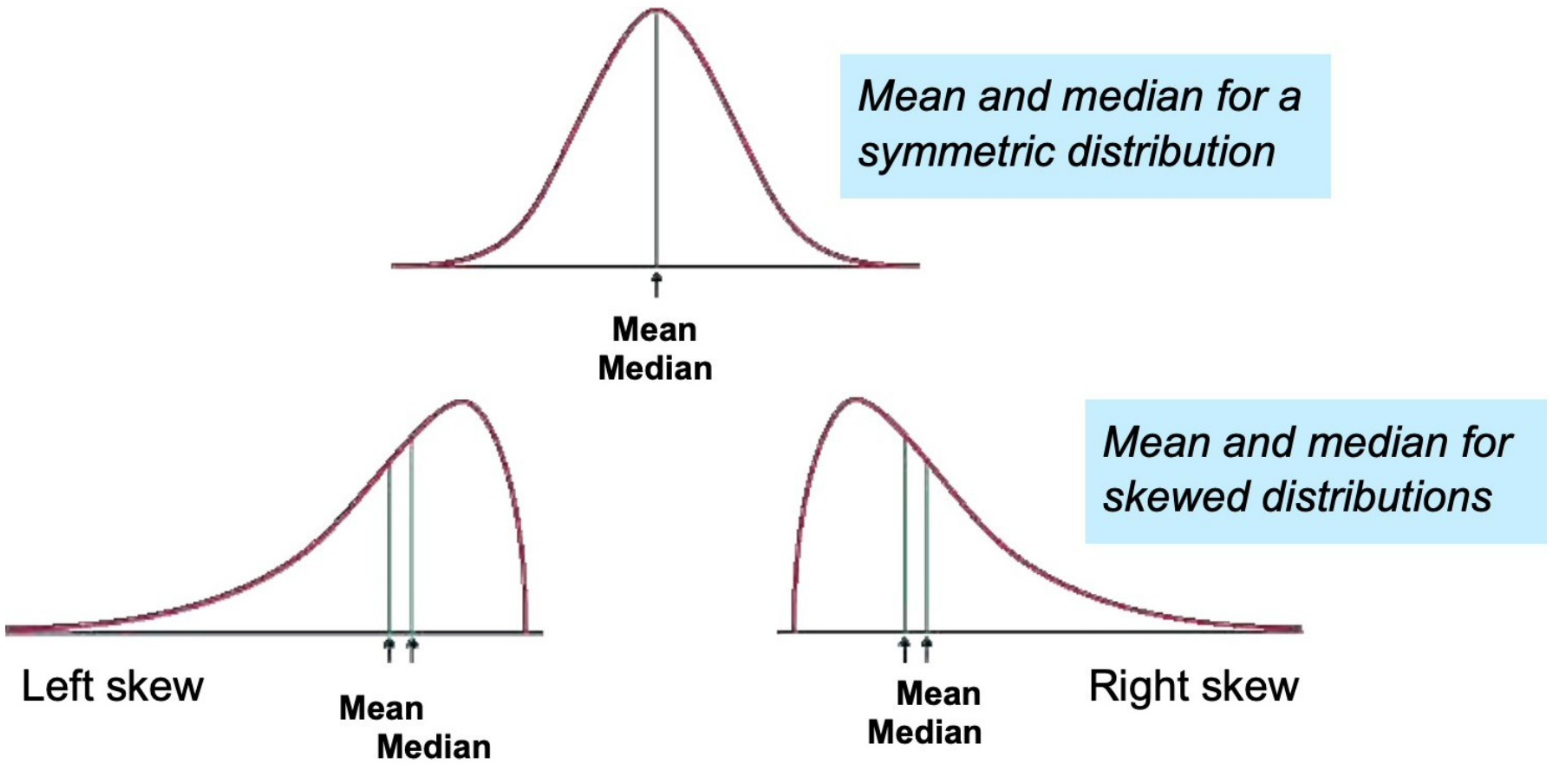
(b)



(c)

Mean vs. Median

- *In a symmetric distribution, mean ≈ median.*
- *If exactly symmetric, then mean = median.*
- *In a skewed distribution, the mean is pulled toward the longer tail.*



Mode

- *It is the most frequently occurring value among all the observations*
- *There may be more than one mode (Unimodal, Bimodal, Trimodal...)*
- *In the previous question:*

  There is no mode because all values occur exactly once

Sample of admission white-blood counts (× 1000) for all patients entering a hospital in Allentown, PA, on a given day

i	x_i	i	x_i
1	7	6	3
2	35	7	10
3	5	8	12
4	9	9	8
5	8		

Mode: 8000

Mean, Median, Mode

Python — vi list_stats.py — 71×42

```python
#!/Library/Frameworks/Python.framework/Versions/3.7/bin/python3

def mean(L):
	total=0
	for i in range(len(L)):
		total=total+L[i]
	mean=total/len(L)
	print("Mean: ",mean)
def median(L):
	L.sort()
	l=len(L)
	if l%2==0:
		median=(L[int(l/2)]+L[int((l-1)/2)])/2
	else:
		median=L[int((l-1)/2)]
	print("Median: ", median)
def mode(L):
	uniq=[]
	for i in range(len(L)):
		if L[i] not in uniq:
			uniq.append(L[i])
		else:
			pass
	uniq_count=[]
	for i in range(len(uniq)):
		count=0
		for j in range(len(L)):
			if uniq[i]==L[j]:
				count+=1
		uniq_count.append(count)
	max_num=max(uniq_count)
	k=uniq_count.index(max_num)
	mode=uniq[k]
	print("Mode: ", mode)

list1=[9,27,18,47,27,9,81,18,27]
list1.sort()
print(list1)
mean(list1)
median(list1)
mode(list1)
```

Python — -bash — 59×7

```
(base) Deepaks-MacBook-Pro:Python deepak$ ./list_stats.py
[9, 9, 18, 18, 27, 27, 27, 47, 81]
Mean:  29.22222222222222
Median:  27
Mode:  27
(base) Deepaks-MacBook-Pro:Python deepak$
```

Delete 27, Mode 9 (erroneous)

Mean, Median, Mode

```
Python — vi stats.py — 47×19
#!/Users/deepak/opt/anaconda3/bin/python3

#####################################
#Statistics
#####################################

import statistics

l=[9,27,18,47,27,9,81,18,27]
mean = statistics.mean(l)
median = statistics.median(l)
mode = statistics.mode(l)

print("The mean is: ",mean)
print("The median is: ",median)
print("The mode is: ",mode)
~
~
"stats.py" 16L, 352C
```

```
Python — -bash — 54×7
(base) Deepaks-MacBook-Pro:Python deepak$ ./stats.py
The mean is:  29.22222222222222
The median is:  27
The mode is:  27
(base) Deepaks-MacBook-Pro:Python deepak$
```

Delete 27

Error: No unique mode (3 equal values)

??

Geometric Mean

- *Useful whenever the quantities to be averaged combine multiplicatively*

The **geometric mean** is the antilogarithm of $\overline{\log x}$, where

$$\overline{\log x} = \frac{1}{n}\sum_{i=1}^{n} \log x_i$$

2 x 8 cm

4 x 4 cm

GM: 4 cm

Measures of variability

The **range** is the difference between the largest and smallest observations in a sample.

The **sample variance**, or **variance**, is defined as follows:

$$s^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}$$

The **sample standard deviation**, or **standard deviation**, is defined as follows:

$$s = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}} = \sqrt{\text{sample variance}}$$

- *In statistics, the number of degrees of freedom is the number of values in the final calculation of a statistic that are free to vary.*
- *For instance, in the above calculations, the degrees of freedom is equal to the number of independent scores (N) minus the number of parameters estimated as intermediate steps (one, namely, the sample mean) and is therefore equal to N–1.*

Quartiles, IQR, Five-number summary

- Quartiles divide data into 4 even parts:
  - First quartile Q1 = 25^{th} percentile
    - 25% of data fall below it and 75% above it
  - Second quartile Q2 = median = 50^{th} percentile
  - Third quartile Q3 = 75^{th} percentile
    - 75% of data fall below it and 25% above it
- Interquartile Range (IQR) = Q3 – Q1
- Five-Number Summary: min, Q1, Median, Q3, max
- Outlier: more than $1.5 \times$ IQR below Q1 and above Q3

Example

- *For the 9 numbers: 43, 35, 43, 33, 38, 53, 64, 27, 34*

$$
\begin{array}{ccc}
43 & & 27 \\
35 & & 33 \\
43 & & 34 \\
33 & & 35 \\
38 & \xrightarrow{\text{sort}} & 38 \\
53 & & 43 \\
64 & & 43 \\
27 & & 53 \\
34 & & 64
\end{array}
$$

$$\left.\begin{array}{c}27\\33\\34\\35\end{array}\right\} \leftarrow \text{median of this half} = \frac{33+34}{2} = 33.5 = Q_1$$

$$38 \leftarrow \text{overall median} = Q_2$$

$$\left.\begin{array}{c}43\\43\\53\\64\end{array}\right\} \leftarrow \text{median of this half} = \frac{43+53}{2} = 48 = Q_3$$

Example

- *For the 9 numbers: 43, 35, 43, 33, 38, 53, 64, 27, 34*

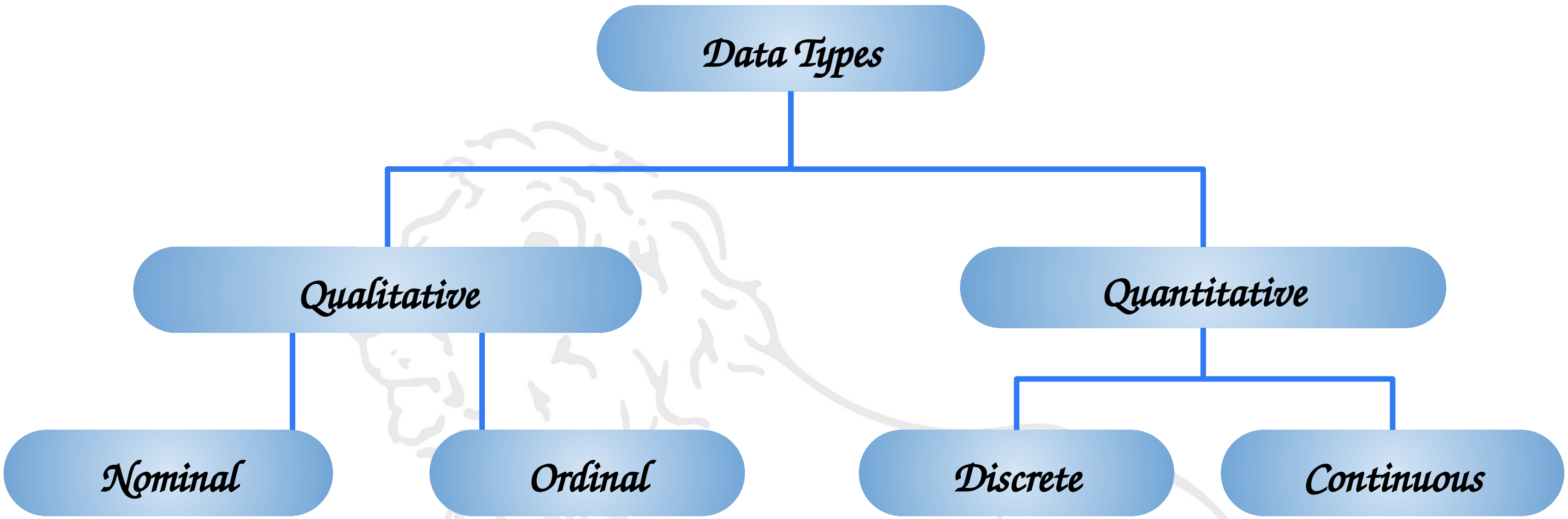
43		27
35		33
43		34
33		35
38	sort $\longrightarrow$	38
53		43
64		43
27		53
34		64

IQR = Q3 – Q1 = 48 – 33.5 = 14.5

Five number summary: 27, 33.5, 38, 48, 64

Are there any outliers?

Data Types


Data Types
Qualitative
Quantitative
Nominal
Ordinal
Discrete
Continuous

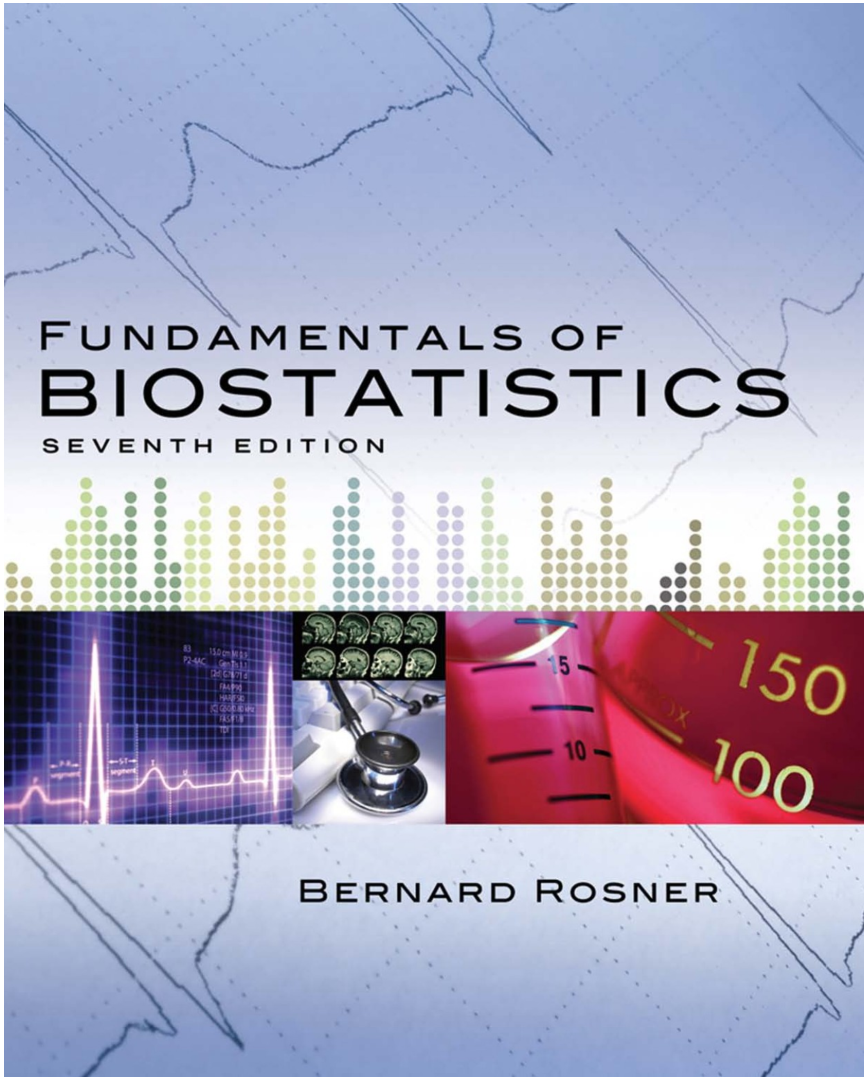
Qualitative data

- *Qualitative or categorical data:* tells the features of the data and categorizes the data into various categories.
  - *Nominal*
    - Consists of categories or names that cannot be ordered or ranked.
    - Often used to categorize observations into groups, and the groups are not comparable.
    - Eg. gender (Male or female), race (White, Black, Asian), religion (Hinduism, Christianity, Islam, Jainism), blood type (A, B, AB, O), etc.
  - *Ordinal*
    - Consists of categories that can be ordered or ranked.
    - However, the distance between categories is not necessarily equal.
    - Eg. education level (Elementary, Middle, High School, College), job position (Manager, Supervisor, Employee), etc.

Quantitative data

- *Quantitative data:* has numerical values.
  - *Discrete*
    - Has discrete values (whole numbers, integers).
    - Eg. number of students, runs scored, etc.
  - *Continuous*
    - Represents the data in a continuous range.
    - Eg. temperature range, salary, etc.

Book



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